```
-- ----------------------------
-- Table: modules
-- ----------------------------
CREATE TABLE IF NOT EXISTS modules (
    id SERIAL PRIMARY KEY,
    name VARCHAR(255) UNIQUE NOT NULL
);

INSERT INTO modules (name) VALUES
('Signal'),
('Diagnosis'),
('Advisory'),
('Contextual'),
('Structural'),
('Scoring'),
('Dependency'),
('Signal / Diagnosis'),
('Diagnosis / Advisory'),
('Advisory / Contextual'),
('Diagnosis / Contextual'),
('Signal / Advisory'),
('Signal / Diagnosis / Advisory'),
('Signal / Diagnosis / Structural'),
('Diagnosis / Structural'),
('Signal / Contextual Guidance'),
('Advisory / Scoring'),
('Diagnosis / Contextual Guidance'),
('Signal / Scoring'),
('Dependency / Contextual Guidance');

-- ----------------------------
-- Table: knowledge_bases
-- ----------------------------
CREATE TABLE IF NOT EXISTS knowledge_bases (
    id SERIAL PRIMARY KEY,
    name VARCHAR(255) UNIQUE NOT NULL,
    description TEXT
);

INSERT INTO knowledge_bases (name, description) VALUES
('SHRM', 'Society for Human Resource Management, HR best practices and frameworks'),
('PMBOK', 'Project Management Body of Knowledge, PMI standards for project management'),
('Lean', 'Lean Management and continuous improvement methodologies'),
('Six Sigma', 'Process improvement and quality management methodologies'),
```

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('Balanced Scorecard', 'Kaplan & Norton strategic performance framework'),
('TOC', 'Theory of Constraints for operational bottleneck management'),
('OKR', 'Objectives and Key Results framework for goal alignment'),
('Porter Five Forces', 'Competitive analysis framework'),
('SWOT', 'Strengths, Weaknesses, Opportunities, Threats analysis'),
('ADKAR', 'Change management framework by Prosci'),
('Design Thinking', 'Human-centered design and innovation methodology'),
('Horizon 1-3 Innovation', 'Innovation portfolio management across three horizons'),
('Blue Ocean Strategy', 'Market differentiation and strategic innovation'),
('Disruptive Innovation Model', 'Framework for assessing disruptive innovations'),
('TRIZ', 'Theory of inventive problem solving'),
('CMMI', 'Capability Maturity Model Integration'),
('ITIL', 'IT service management best practices'),
('COBIT', 'IT governance framework'),
('EFQM', 'European Foundation for Quality Management excellence model'),
('Malcolm Baldrige', 'US national quality framework for organizational excellence');

-- ----------------------------
-- Table: frameworks
-- ----------------------------
CREATE TABLE IF NOT EXISTS frameworks (
    id SERIAL PRIMARY KEY,
    name VARCHAR(255) NOT NULL,
    primary_module_id INT REFERENCES modules(id),
    status_relevance VARCHAR(255),
    temporal_context VARCHAR(255),
    dependencies_links TEXT,
    notes_historical_context TEXT
);

-- First batch of 200 frameworks
INSERT INTO frameworks (name, primary_module_id, status_relevance, temporal_context,
dependencies_links, notes_historical_context) VALUES
('Balanced Scorecard', (SELECT id FROM modules WHERE name='Signal / Scoring'), 'All',
'Monthly / Quarterly', 'OKR Alignment, Strategic initiatives', 'Past KPIs inform priority scoring'),
('Lean', (SELECT id FROM modules WHERE name='Signal / Diagnosis / Structural'), 'Pending /
Ongoing', 'Daily / Weekly', 'TOC, Kaizen', 'Past process improvements feed Contextual
guidance'),
('Theory of Constraints', (SELECT id FROM modules WHERE name='Signal / Diagnosis /
Dependency'), 'All', 'Initiative timing', 'Critical Chain PM', 'Previous bottlenecks inform
prioritization'),
('SWOT Analysis', (SELECT id FROM modules WHERE name='Diagnosis'), 'New', 'Initiative
launch', NULL, 'Prior SWOTs inform recommendations'),
```

('Porter Five Forces', (SELECT id FROM modules WHERE name='Diagnosis'), 'New', 'Initiative launch', NULL, 'Prior SWOTs inform recommendations'),
('5 Whys / Fishbone Diagram', (SELECT id FROM modules WHERE name='Diagnosis'), 'Pending / Completed', 'Event-driven', 'Lean / TOC', 'Historical root causes guide Contextual layer'),
('OKR Alignment', (SELECT id FROM modules WHERE name='Advisory / Contextual'), 'New / Pending', 'Quarterly', 'Balanced Scorecard, Strategic initiatives', 'Past OKR success/failure informs INTJ-style advice'),
('Eisenhower Matrix', (SELECT id FROM modules WHERE name='Advisory / Contextual'), 'New', 'Daily / Weekly', 'Initiative prioritization', 'Historical delays influence urgency'),
('Cynefin Framework', (SELECT id FROM modules WHERE name='Contextual Guidance'), 'All', 'Initiative lifecycle', 'Diagnosis, Advisory', 'Helps anticipate complex vs chaotic contexts'),
('Scenario Planning', (SELECT id FROM modules WHERE name='Advisory / Contextual'), 'New / Pending', 'Project / Quarterly', 'OKR, Balanced Scorecard', 'Historical scenario success informs weighting'),
('Critical Chain Project Management', (SELECT id FROM modules WHERE name='Signal / Contextual'), 'Pending', 'Initiative deadlines', 'TOC, Dependency mapping', 'Past timing adjustments guide buffer calculations'),
('Root Cause Analysis (DMAIC, Lean Six Sigma)', (SELECT id FROM modules WHERE name='Diagnosis'), 'Pending', 'Event-triggered', 'Lean / Six Sigma', 'Historical fixes inform Contextual layer'),
('Value Stream Mapping', (SELECT id FROM modules WHERE name='Diagnosis / Structural'), 'Pending', 'Continuous improvement', 'Lean, TOC', 'Past flow changes guide advisory'),
('Influence & Persuasion Principles (Cialdini)', (SELECT id FROM modules WHERE name='Advisory'), 'New / Pending', 'Stakeholder engagement', 'OKR, Initiative execution', 'Prior communications outcomes guide approach'),
('Monte Carlo Simulation', (SELECT id FROM modules WHERE name='Contextual Guidance'), 'New / Pending', 'Initiative risk', 'Dependency scoring', 'Prior simulations inform probability weighting'),
('Dependency Mapping / PERT Analysis', (SELECT id FROM modules WHERE name='Dependency'), 'All', 'Initiative & timeline', 'Critical Chain / TOC', 'Past conflicts inform risk scoring'),
('Galbraith Star Model', (SELECT id FROM modules WHERE name='Structural'), 'All', 'Quarterly / Initiative', 'OKRs, Balanced Scorecard', 'Past structural fixes inform recommendations'),
('Hoshin Kanri', (SELECT id FROM modules WHERE name='Structural'), 'Pending', 'Quarterly / Annual', 'Strategic initiatives', 'Prior planning cycles inform alignment'),
('McKinsey 7S Framework', (SELECT id FROM modules WHERE name='Diagnosis / Structural'), 'All', 'Quarterly', 'Balanced Scorecard', 'Historical org misalignment informs Contextual guidance'),
('RACI Matrix', (SELECT id FROM modules WHERE name='Signal / Diagnosis'), 'New', 'Initiative-driven', 'Project management', 'Previous accountability gaps guide assignment'),
('Portfolio Prioritization Matrix', (SELECT id FROM modules WHERE name='Advisory'), 'New / Pending', 'Quarterly', 'OKRs, Balanced Scorecard', 'Historical success of initiatives informs ranking'),

('MoSCoW Prioritization', (SELECT id FROM modules WHERE name='Advisory'), 'New', 'Initiative', 'Portfolio matrix', 'Past prioritization informs context-aware recommendations'),
('Stage-Gate Process', (SELECT id FROM modules WHERE name='Advisory'), 'Pending', 'Product launch', 'Portfolio prioritization', 'Historical gating outcomes inform timing'),
('Scenario Analysis', (SELECT id FROM modules WHERE name='Contextual Guidance'), 'New / Pending', 'Initiative / Risk', 'Advisory frameworks', 'Prior scenario results adjust recommendations'),
('Strategic Choice Cascade', (SELECT id FROM modules WHERE name='Diagnosis / Advisory'), 'New', 'Quarterly', 'OKR Alignment', 'Past strategic choices feed Contextual guidance'),
('VRIO Framework', (SELECT id FROM modules WHERE name='Diagnosis'), 'New', 'Initiative launch', 'SWOT, Porter', 'Historical capability analysis informs strategy'),
('PESTEL Analysis', (SELECT id FROM modules WHERE name='Diagnosis'), 'New', 'Initiative / Annual', 'SWOT', 'Prior environmental scan informs recommendations'),
('BCG Growth-Share Matrix', (SELECT id FROM modules WHERE name='Diagnosis / Advisory'), 'Pending', 'Quarterly', 'Portfolio prioritization', 'Historical market share shifts inform decisions'),
('GE/McKinsey Matrix', (SELECT id FROM modules WHERE name='Diagnosis / Advisory'), 'Pending', 'Quarterly', 'Portfolio prioritization', 'Past assessment informs risk/reward'),
('Blue Ocean Strategy', (SELECT id FROM modules WHERE name='Diagnosis / Advisory'), 'New', 'Initiative', 'Scenario planning', 'Past innovation projects inform recommendations'),
('Jobs to Be Done Framework', (SELECT id FROM modules WHERE name='Advisory'), 'New', 'Initiative', 'Marketing, Product', 'Past implementation informs Contextual guidance'),
('Value Proposition Canvas', (SELECT id FROM modules WHERE name='Advisory'), 'New', 'Initiative', 'Marketing, Product', 'Prior product-market fits inform prioritization'),
('Stakeholder Mapping', (SELECT id FROM modules WHERE name='Signal / Advisory'), 'New / Pending', 'Project kickoff', 'RACI, Influence Principles', 'Historical stakeholder engagement informs guidance'),
('Customer Journey Mapping', (SELECT id FROM modules WHERE name='Diagnosis / Advisory'), 'New', 'Quarterly', 'Marketing, Product', 'Past journey improvements inform prioritization'),
('AIDA / Pirate Metrics (AARRR)', (SELECT id FROM modules WHERE name='Advisory'), 'New / Pending', 'Marketing cycle', 'OKR, Initiative', 'Historical campaigns inform prioritization'),
('Conversion Funnel Analysis', (SELECT id FROM modules WHERE name='Advisory'), 'New', 'Marketing / Initiative', 'Metrics, KPIs', 'Past funnel outcomes guide timing & focus'),
('Net Promoter Score', (SELECT id FROM modules WHERE name='Signal / Scoring'), 'All', 'Monthly', 'Customer Experience KPIs', 'Historical NPS changes guide advisory actions'),
('Customer Experience Maturity Model', (SELECT id FROM modules WHERE name='Diagnosis / Scoring'), 'Pending', 'Quarterly', 'Balanced Scorecard', 'Historical experience scores inform INTJ guidance'),
('Brand Archetype Framework', (SELECT id FROM modules WHERE name='Advisory'), 'New', 'Marketing / Branding', 'Stakeholder mapping', 'Historical branding informs messaging'),
('Marketing Mix / 4Ps', (SELECT id FROM modules WHERE name='Advisory'), 'New', 'Marketing cycle', 'OKR / Initiative', 'Historical mix changes inform recommendation'),

('STP Model', (SELECT id FROM modules WHERE name='Diagnosis / Advisory'), 'New', 'Marketing cycle', 'Marketing, Portfolio', 'Historical segmentation informs prioritization'),
('Horizon 1-3 Innovation Model', (SELECT id FROM modules WHERE name='Advisory / Contextual'), 'New / Pending', 'Quarterly', 'Portfolio matrix', 'Past innovations inform timing & priority'),
('Innovation Ambition Matrix', (SELECT id FROM modules WHERE name='Advisory / Contextual'), 'New', 'Initiative', 'Horizon 1-3, Scenario Planning', 'Past initiatives inform weighting'),
('Disruptive Innovation Model', (SELECT id FROM modules WHERE name='Diagnosis / Advisory'), 'New', 'Initiative', 'Scenario planning', 'Historical disruption informs prioritization'),
('Product-Market Fit Pyramid', (SELECT id FROM modules WHERE name='Advisory / Contextual'), 'New', 'Initiative', 'OKR, Balanced Scorecard', 'Past product successes guide recommendations'),
('OKR Cascading Framework', (SELECT id FROM modules WHERE name='Advisory / Contextual'), 'New / Pending', 'Quarterly', 'Balanced Scorecard', 'Previous OKR cascades inform timing'),
('Strategic Risk Assessment', (SELECT id FROM modules WHERE name='Diagnosis'), 'Pending', 'Initiative / Quarterly', 'Risk Heat Map', 'Historical risk events inform Contextual layer');
---- ----------
-- Frameworks Batch 2 (61-120)
-- ----------
INSERT INTO frameworks (name, primary_module, status_relevance, temporal_context, dependencies_links, notes_historical_context)
VALUES
('Lean Six Sigma', 'Signal / Diagnosis / Structural', 'Pending', 'Daily / Weekly', 'Lean, Six Sigma, TOC', 'Prior process improvements inform advisory'),
('DMAIC Cycle', 'Diagnosis', 'Pending', 'Event-driven', 'Lean Six Sigma', 'Past DMAIC cycles guide Contextual recommendations'),
('Kaizen / Continuous Improvement', 'Diagnosis / Structural', 'Ongoing', 'Daily / Weekly', 'Lean', 'Historical improvements influence prioritization'),
('PDCA Cycle', 'Diagnosis / Structural', 'Pending', 'Daily / Weekly', 'Lean, Continuous Improvement', 'Previous PDCA iterations inform recommendations'),
('Total Quality Management (TQM)', 'Diagnosis / Structural', 'All', 'Quarterly', 'Lean, KPI Scoring', 'Historical TQM metrics guide scoring'),
('Just-in-Time (JIT) Production', 'Signal / Diagnosis', 'Pending', 'Daily / Weekly', 'Lean, Kanban', 'Historical JIT successes guide resource allocation'),
('Kanban', 'Signal / Diagnosis', 'Pending', 'Daily / Weekly', 'Lean, JIT', 'Past WIP limits inform recommendations'),
('Scrum / Agile Framework', 'Signal / Diagnosis / Advisory', 'Pending', 'Sprint cycles', 'Kanban, Lean', 'Previous sprints inform velocity & backlog prioritization'),
('Critical Path Method (CPM)', 'Dependency / Contextual Guidance', 'Pending', 'Initiative deadlines', 'PERT, TOC', 'Historical delays inform buffer planning'),
('PERT Analysis', 'Dependency / Contextual Guidance', 'Pending', 'Initiative', 'CPM, Critical Chain', 'Past PERT adjustments inform scheduling recommendations'),

('Bottleneck Analysis', 'Signal / Diagnosis', 'Pending', 'Daily / Weekly', 'TOC, Lean', 'Historical bottlenecks inform Contextual prioritization'),
('SIPOC Mapping', 'Diagnosis / Structural', 'New / Pending', 'Initiative process design', 'Lean, Value Stream Mapping', 'Historical processes guide advisory layer'),
('Business Process Reengineering (BPR)', 'Structural', 'Pending', 'Quarterly / Initiative', 'Lean, Six Sigma', 'Prior BPR outcomes inform future structural remedies'),
('Process Capability Analysis (Cp, Cpk)', 'Scoring / Diagnosis', 'Pending', 'Continuous / Monthly', 'Lean, TQM', 'Historical capability informs maturity scoring'),
('Failure Mode & Effects Analysis (FMEA)', 'Diagnosis / Contextual Guidance', 'Pending', 'Initiative', 'Risk Scoring, TOC', 'Past failures inform INTJ-style recommendations'),
('Service Blueprinting', 'Diagnosis / Advisory', 'New', 'Initiative / Quarterly', 'Customer Journey Mapping', 'Past service redesigns inform advisory actions'),
('Work Breakdown Structure (WBS)', 'Signal / Advisory', 'New / Pending', 'Project kickoff', 'CPM, Critical Chain', 'Historical WBS informs Contextual planning'),
('Resource Loading & Leveling', 'Signal / Contextual Guidance', 'Pending', 'Daily / Weekly', 'Dependency, Critical Chain', 'Historical resource allocation informs recommendations'),
('Demand-Capacity Balancing', 'Signal / Contextual Guidance', 'Pending', 'Daily / Weekly', 'Resource Loading, TOC', 'Historical capacity informs advisory layer'),
('Time-Driven Activity-Based Costing (TDABC)', 'Scoring', 'Pending', 'Monthly / Quarterly', 'Financial KPIs', 'Prior cost allocations inform Contextual prioritization'),
('PMBOK Knowledge Areas', 'Signal / Diagnosis / Advisory', 'All', 'Project lifecycle', 'WBS, RACI, Portfolio Matrix', 'Past projects inform Contextual recommendations'),
('PRINCE2 Methodology', 'Signal / Diagnosis / Advisory', 'Pending', 'Project', 'Stage-Gate, Portfolio Prioritization', 'Historical project performance informs advisory layer'),
('Stage-Gate Process', 'Advisory', 'Pending', 'Product launch', 'Portfolio Prioritization, Scenario Planning', 'Past stage outcomes inform Contextual recommendations'),
('RAID Log (Risks, Assumptions, Issues, Dependencies)', 'Signal / Dependency', 'Pending', 'Initiative', 'Risk Heat Map, Dependency Mapping', 'Historical logs inform advisory guidance'),
('RACI Matrix (Project Application)', 'Signal / Diagnosis', 'New / Pending', 'Initiative', 'Stakeholder Mapping', 'Historical accountability informs Contextual recommendations'),
('MoSCoW Prioritization', 'Advisory', 'New', 'Initiative', 'Portfolio Matrix', 'Prior prioritizations inform timing & urgency'),
('Eisenhower Matrix', 'Advisory / Contextual Guidance', 'New', 'Daily / Weekly', 'Portfolio Prioritization', 'Past urgency assessments inform INTJ prioritization'),
('Decision Matrix / Weighted Scoring', 'Advisory / Contextual', 'New / Pending', 'Initiative', 'Portfolio / OKR', 'Historical scoring informs priority recommendations'),
('Force Field Analysis', 'Diagnosis / Advisory', 'Pending', 'Initiative', 'Scenario Planning', 'Historical forces inform recommendation weighting'),
('Gap Analysis', 'Diagnosis', 'Pending', 'Quarterly / Initiative', 'SWOT, Balanced Scorecard', 'Prior gaps inform advisory recommendations'),
('Sensitivity Analysis', 'Contextual Guidance', 'Pending', 'Initiative / Quarterly', 'Monte Carlo Simulation', 'Historical variability informs recommendations'),
('Pareto Analysis (80/20 Rule)', 'Diagnosis / Advisory', 'New / Pending', 'Initiative / Continuous', 'KPI Scoring', 'Historical analysis informs priority actions'),

('Ladder of Inference', 'Contextual Guidance', 'New / Pending', 'Decision-making', 'Diagnosis, Advisory', 'Past inferences guide decision logic'),
('Red Team / Blue Team Analysis', 'Contextual Guidance', 'Pending', 'Initiative / Risk Assessment', 'Scenario Planning', 'Historical risk testing informs advisory recommendations'),
('Premortem Analysis', 'Contextual Guidance', 'Pending', 'Initiative / Project launch', 'Advisory', 'Past project failures inform Contextual recommendations'),
('Influence Diagram', 'Contextual Guidance', 'New / Pending', 'Initiative / Decision-making', 'Dependency Analysis', 'Historical influences inform recommendation weighting'),
('Causal Loop Diagram', 'Contextual Guidance', 'Pending', 'Continuous / Project', 'Dependency Analysis', 'Prior loops guide advisory recommendations'),
('Systems Thinking Loops', 'Contextual Guidance', 'Pending', 'Continuous / Quarterly', 'Dependency, Diagnosis', 'Historical system interactions inform Contextual guidance'),
('Heuristics & Bias Checklist', 'Contextual Guidance', 'New', 'Initiative / Decision', 'Advisory, Diagnosis', 'Historical biases inform recommendations'),
('Decision Fatigue Awareness', 'Contextual Guidance', 'All', 'Daily / Weekly', 'Advisory', 'Prior fatigue patterns inform INTJ guidance'),
('Transactional Analysis', 'Contextual Guidance', 'Pending', 'Initiative / Stakeholder', 'Advisory, Influence Principles', 'Prior interactions inform recommendations'),
('Stress-Performance Curve (Yerkes-Dodson)', 'Contextual Guidance', 'All', 'Daily / Weekly', 'Advisory', 'Past performance under stress informs Contextual prioritization'),
('Flow State Model (Csikszentmihalyi)', 'Contextual Guidance', 'All', 'Daily / Weekly', 'Advisory', 'Historical flow insights inform recommendation timing'),
('Cognitive Bias Codex', 'Contextual Guidance', 'New / Pending', 'Decision-making', 'Heuristics, Advisory', 'Historical bias incidents inform advisory weighting'),
('Self-Determination Theory', 'Contextual Guidance', 'Pending', 'Initiative / Team', 'Leadership Models', 'Prior motivation patterns inform INTJ recommendations'),
('Emotional Intelligence (Goleman)', 'Advisory / Contextual', 'All', 'Daily / Weekly', 'Leadership Models', 'Historical EI outcomes inform recommendations'),
('Situational Leadership (Hersey-Blanchard)', 'Advisory / Contextual', 'Pending', 'Team management', 'Leadership Pipeline', 'Past leadership assessments guide Contextual guidance'),
('Servant Leadership Model', 'Advisory / Contextual', 'Pending', 'Team management', 'EI, Leadership Pipeline', 'Historical team outcomes inform advisory style'),
('Kotter’s 8-Step Change Model', 'Diagnosis / Contextual Guidance', 'Pending', 'Change initiatives', 'Lewin Change Model, ADKAR', 'Past change initiatives inform recommendations'),
('Lewin’s Change Model', 'Diagnosis / Contextual Guidance', 'Pending', 'Change initiatives', 'Kotter, ADKAR', 'Prior interventions guide advisory layer'),
('ADKAR Model', 'Diagnosis / Contextual Guidance', 'Pending', 'Change initiatives', 'Lewin, Kotter', 'Historical adoption informs Contextual recommendations'),
('Tuckman Team Development Model', 'Advisory / Contextual', 'All', 'Team / Quarterly', 'Leadership, EI', 'Historical team development stages inform advisory'),
('High-Performance Organization Model', 'Advisory / Structural', 'Pending', 'Quarterly', 'Balanced Scorecard, Leadership Models', 'Historical performance informs scoring & recommendations'),
('Talent Management Maturity Model', 'Scoring / Contextual', 'Pending', 'Quarterly', 'HR KPIs, OKR', 'Prior talent development informs Contextual advisory'),

('Leadership Pipeline', 'Advisory / Contextual', 'Pending', 'Quarterly / Initiative', 'Leadership Models', 'Historical succession patterns inform recommendations'),
('Cultural Web Model', 'Diagnosis / Contextual Guidance', 'Pending', 'Quarterly', 'Leadership, Org Design', 'Past culture initiatives inform advisory recommendations'),
('Decision Rights Matrix', 'Advisory / Contextual', 'New', 'Initiative', 'RACI, Portfolio', 'Historical decision rights inform advisory'),
('Span of Control Analysis', 'Diagnosis / Scoring', 'All', 'Quarterly / Annual', 'Leadership, Org Design', 'Historical control spans inform INTJ prioritization');
-- ----------
-- Frameworks Batch 3 (121-200)
-- ----------
INSERT INTO frameworks (name, primary_module, status_relevance, temporal_context, dependencies_links, notes_historical_context)
VALUES
('Maslow's Hierarchy of Needs', 'Contextual Guidance', 'All', 'Team / Initiative', 'Leadership, EI', 'Historical motivation informs advisory timing'),
('Herzberg Motivation-Hygiene', 'Contextual Guidance', 'Pending', 'Team / Initiative', 'Leadership Models', 'Past motivational interventions guide recommendations'),
('McClelland's Theory of Needs', 'Contextual Guidance', 'Pending', 'Team / Initiative', 'Leadership, Talent Management', 'Historical performance outcomes inform prioritization'),
('Self-Determination Theory', 'Contextual Guidance', 'Pending', 'Initiative / Team', 'Leadership, EI', 'Prior motivational strategies inform advisory'),
('Big Five Personality Traits (OCEAN)', 'Contextual Guidance', 'Pending', 'Team / Individual', 'Leadership, HR', 'Historical team composition informs Contextual recommendations'),
('DISC Assessment', 'Advisory / Contextual', 'Pending', 'Team / Initiative', 'Leadership, EI', 'Prior behavior patterns guide INTJ advisory'),
('Emotional Intelligence (Goleman)', 'Advisory / Contextual', 'All', 'Daily / Weekly', 'Leadership Models', 'Historical EI insights inform team recommendations'),
('Cognitive Load Theory', 'Contextual Guidance', 'Pending', 'Project / Team', 'Decision Fatigue, Advisory', 'Historical overload informs INTJ prioritization'),
('Behavioral Economics Principles', 'Contextual Guidance', 'New / Pending', 'Initiative / Decision', 'Heuristics, Bias Codex', 'Prior behavioral patterns inform recommendations'),
('Nudge Theory', 'Contextual Guidance', 'New / Pending', 'Initiative / Team', 'Behavioral Economics', 'Past nudges inform advisory strategies'),
('Loss Aversion / Prospect Theory', 'Contextual Guidance', 'New / Pending', 'Initiative / Risk', 'Scenario Planning', 'Historical losses inform INTJ risk weighting'),
('Anchoring & Adjustment Bias', 'Contextual Guidance', 'Pending', 'Decision-making', 'Heuristics', 'Past decision errors inform Contextual guidance'),
('Availability Heuristic', 'Contextual Guidance', 'Pending', 'Decision-making', 'Heuristics', 'Historical outcomes inform advisory weighting'),
('Representativeness Heuristic', 'Contextual Guidance', 'Pending', 'Decision-making', 'Heuristics', 'Prior misclassifications inform advisory'),
('Sunk Cost Fallacy Awareness', 'Contextual Guidance', 'Pending', 'Initiative / Decision', 'Advisory, Portfolio', 'Historical sunk cost patterns guide recommendations'),

('Confirmation Bias Awareness', 'Contextual Guidance', 'Pending', 'Decision-making', 'Heuristics', 'Past decisions inform advisory weighting'),
('Framing Effect Awareness', 'Contextual Guidance', 'Pending', 'Initiative / Decision', 'Behavioral Economics', 'Historical framing informs recommendation approach'),
('Anchoring / Adjustment in Negotiation', 'Advisory / Contextual', 'Pending', 'Stakeholder engagement', 'Influence Principles', 'Past negotiation outcomes inform advisory'),
('Social Proof & Bandwagon Effect', 'Contextual Guidance', 'Pending', 'Team / Initiative', 'Influence Principles', 'Historical peer adoption informs recommendations'),
('Scarcity Principle', 'Advisory / Contextual', 'New / Pending', 'Initiative / Resource allocation', 'Behavioral Economics', 'Historical urgency signals inform advisory timing'),
('Habit Loop (Cue-Routine-Reward)', 'Contextual Guidance', 'Pending', 'Team / Initiative', 'Behavioral Interventions', 'Past behavior informs advisory layer'),
('Fogg Behavior Model', 'Contextual Guidance', 'Pending', 'Team / Initiative', 'Habit Loop', 'Historical adherence informs recommendations'),
('COM-B Behavior Model', 'Contextual Guidance', 'Pending', 'Initiative / Change', 'Fogg Model, Behavioral Interventions', 'Prior behavior informs INTJ advisory'),
('ADKAR Model', 'Diagnosis / Contextual Guidance', 'Pending', 'Change initiatives', 'Lewin, Kotter', 'Historical adoption patterns inform advisory'),
('Gantt Charts', 'Signal / Dependency / Advisory', 'Pending', 'Project timeline', 'CPM, Critical Chain', 'Historical schedule adherence informs recommendations'),
('Monte Carlo Simulation', 'Contextual Guidance', 'Pending', 'Initiative / Risk', 'Scenario Planning', 'Past probabilistic outcomes inform advisory'),
('Decision Tree Analysis', 'Contextual Guidance', 'New / Pending', 'Initiative / Risk', 'Sensitivity Analysis', 'Historical decisions inform weighting & probability'),
('Bayesian Updating', 'Contextual Guidance', 'Pending', 'Decision-making', 'Decision Trees', 'Past observations inform probability adjustments'),
('Risk Heat Map (ISO 31000)', 'Signal / Diagnosis', 'All', 'Initiative / Quarterly', 'Dependency Mapping', 'Past risks inform Contextual prioritization'),
('Enterprise Risk Management (ERM)', 'Diagnosis / Advisory', 'Pending', 'Quarterly / Initiative', 'Risk Heat Map', 'Historical ERM results inform advisory'),
('Failure Mode & Effects Analysis (FMEA)', 'Diagnosis / Contextual Guidance', 'Pending', 'Initiative', 'Risk Scoring', 'Historical failures inform INTJ recommendations'),
('Hazard & Operability Analysis (HAZOP)', 'Diagnosis / Contextual Guidance', 'Pending', 'Initiative / Process', 'Lean, Risk Assessment', 'Past process hazards inform recommendations'),
('Bowtie Risk Analysis', 'Contextual Guidance', 'Pending', 'Initiative / Risk', 'ERM', 'Historical failures inform advisory'),
('COSO ERM Framework', 'Diagnosis / Advisory', 'Pending', 'Quarterly', 'Risk Heat Map', 'Historical internal controls inform scoring'),
('ISO 31000 Risk Management', 'Signal / Diagnosis / Contextual', 'All', 'Initiative / Quarterly', 'ERM', 'Historical risk controls inform advisory'),
('Basel III / Banking Risk Framework', 'Contextual Guidance', 'Pending', 'Financial Operations', 'ERM', 'Past compliance informs risk recommendations'),
('CAPM (Capital Asset Pricing Model)', 'Contextual Guidance', 'Pending', 'Investment / Initiative', 'Scenario Analysis', 'Historical returns inform weighting recommendations'),

('Modern Portfolio Theory (MPT)', 'Contextual Guidance', 'Pending', 'Portfolio / Initiative', 'CAPM', 'Historical asset allocation informs advisory'),
('Black-Litterman Model', 'Contextual Guidance', 'Pending', 'Portfolio / Initiative', 'MPT, Scenario Planning', 'Historical weights inform recommendations'),
('Risk-Adjusted Return on Capital (RAROC)', 'Scoring / Contextual', 'Pending', 'Quarterly', 'ERM, Financial KPIs', 'Past returns inform scoring'),
('EVA / Economic Value Added', 'Scoring / Contextual', 'Pending', 'Quarterly', 'Financial KPIs', 'Historical performance informs advisory'),
('Balanced Scorecard – Financial KPIs', 'Scoring / Signal', 'All', 'Quarterly', 'OKRs', 'Historical financial performance informs recommendations'),
('Activity-Based Costing (ABC)', 'Scoring', 'Pending', 'Monthly', 'TDABC', 'Historical cost allocation informs advisory'),
('Time-Driven ABC', 'Scoring', 'Pending', 'Monthly', 'Activity-Based Costing', 'Past resource costs inform Contextual guidance'),
('KPI Tree / Dashboard', 'Signal / Scoring', 'All', 'Weekly / Monthly', 'Balanced Scorecard, OKR', 'Historical trends inform recommendations'),
('Leading vs Lagging Indicators', 'Signal / Scoring', 'All', 'Monthly / Quarterly', 'KPI Tree', 'Historical trends inform advisory & prioritization'),
('Benchmarking (Competitive / Internal)', 'Diagnosis / Advisory', 'Pending', 'Quarterly', 'Balanced Scorecard', 'Historical benchmarks inform recommendations'),
('Hoshin Kanri / Policy Deployment', 'Structural / Advisory', 'Pending', 'Quarterly / Annual', 'OKR Alignment', 'Past deployment informs Contextual prioritization'),
('Organizational Network Analysis (ONA)', 'Dependency / Contextual', 'Pending', 'Quarterly / Initiative', 'Stakeholder Mapping', 'Historical interactions inform INTJ advisory'),
('Sociomatrix / Social Graph Analysis', 'Dependency / Contextual', 'Pending', 'Quarterly / Initiative', 'ONA', 'Historical connections inform advisory recommendations'),
('Complexity Theory Applications', 'Contextual Guidance', 'Pending', 'Initiative / System', 'Cynefin, Dependency Mapping', 'Past complexity outcomes inform INTJ recommendations'),
('Systems Dynamics Modeling', 'Contextual Guidance', 'Pending', 'Quarterly / Initiative', 'Causal Loops, Feedback Loops', 'Historical system feedback informs advisory'),
('Feedback Loop Analysis', 'Contextual Guidance', 'Pending', 'Continuous', 'Systems Thinking', 'Prior loop outcomes inform recommendations'),
('Control Charts', 'Signal / Scoring', 'Pending', 'Continuous / Monthly', 'Lean, TQM', 'Historical process control informs recommendations'),
('Statistical Process Control (SPC)', 'Diagnosis / Scoring', 'Pending', 'Continuous', 'Lean, Control Charts', 'Historical deviations inform advisory'),
('Six Sigma DMAIC', 'Diagnosis / Advisory', 'Pending', 'Event-driven / Initiative', 'Lean', 'Historical DMAIC results inform Contextual guidance'),
('Six Sigma DMADV', 'Diagnosis / Advisory', 'Pending', 'Initiative / Product', 'Lean, Design Thinking', 'Past design outcomes inform advisory'),
('Design of Experiments (DOE)', 'Contextual Guidance / Diagnosis', 'Pending', 'Initiative', 'SPC, Lean', 'Historical experiments inform advisory'),
('Taguchi Methods', 'Contextual Guidance', 'Pending', 'Initiative / Product', 'DOE', 'Historical robustness testing informs advisory'),

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('Root Cause Corrective Action (RCCA)', 'Diagnosis / Contextual', 'Pending', 'Post-event',
'FMEA, Lean', 'Historical corrective actions inform recommendations'),
('Corrective & Preventive Actions (CAPA)', 'Structural / Contextual', 'Pending', 'Post-event',
'RCCA', 'Prior CAPA informs INTJ advisory'),
('Quality Function Deployment (QFD)', 'Diagnosis / Advisory', 'Pending', 'Product / Initiative',
'Voice of Customer, DOE', 'Past deployments inform advisory'),
('Kano Model', 'Advisory / Contextual', 'Pending', 'Product / Customer', 'QFD', 'Historical feature
adoption informs recommendations'),
('House of Quality', 'Diagnosis / Advisory', 'Pending', 'Product / Initiative', 'QFD, Kano', 'Prior
mappings inform advisory'),
('Failure Tree Analysis (FTA)', 'Diagnosis / Contextual', 'Pending', 'Initiative / Risk', 'FMEA',
'Historical failure patterns inform advisory'),
('Event Tree Analysis (ETA)', 'Diagnosis / Contextual', 'Pending', 'Initiative / Risk', 'FTA',
'Historical events inform recommendations'),
('Probabilistic Risk Assessment (PRA)', 'Contextual Guidance', 'Pending', 'Initiative / Risk',
'Monte Carlo, ETA', 'Past probabilistic outcomes inform advisory'),
('Decision Quality Framework', 'Contextual Guidance', 'New / Pending', 'Initiative / Decision',
'Decision Trees, Scenario Planning', 'Historical decision outcomes inform INTJ advisory'),
('NPV / Net Present Value', 'Scoring / Contextual', 'Pending', 'Initiative / Investment', 'ROI,
Financial KPIs', 'Historical outcomes inform advisory'),
('IRR / Internal Rate of Return', 'Scoring / Contextual', 'Pending', 'Initiative / Investment', 'NPV',
'Historical returns inform recommendations'),
('Payback Period Analysis', 'Scoring / Contextual', 'Pending', 'Initiative / Investment', 'ROI', 'Past
payback experiences inform advisory'),
('Risk-Adjusted ROI', 'Scoring / Contextual', 'Pending', 'Initiative / Investment', 'NPV, Scenario
Planning', 'Historical adjustments inform advisory'),
('Real Options Analysis', 'Contextual Guidance', 'Pending', 'Initiative / Investment', 'Scenario
Planning', 'Past option exercises inform advisory'),
('Monte Carlo Simulation for Finance', 'Contextual Guidance', 'Pending', 'Investment / Risk',
'Scenario Planning', 'Historical outcomes guide INTJ prioritization');
-- 1. Create the junction table to link frameworks to knowledge bases
CREATE TABLE IF NOT EXISTS framework_knowledge_link (
   id SERIAL PRIMARY KEY,
   framework_id INT NOT NULL REFERENCES frameworks(id) ON DELETE CASCADE,
   knowledge_base_id INT NOT NULL REFERENCES knowledge_bases(id) ON DELETE
CASCADE
);

-- 2. Insert links between frameworks and knowledge bases
-- Note: Adjust the framework_id and knowledge_base_id based on your IDs in your tables.

-- Balanced Scorecard → Strategic Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (1, 1);
```

```
-- Lean → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (2, 2);

-- Theory of Constraints → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (3, 2);

-- SWOT Analysis → Strategic Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (4, 1);

-- 5 Whys / Fishbone Diagram → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (6, 2);

-- OKR Alignment → Strategy & Performance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (7, 3);

-- Eisenhower Matrix → Productivity & Prioritization
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (8, 4);

-- Cynefin Framework → Strategy & Contextual Analysis
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (9, 3);

-- Scenario Planning → Strategic Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (10, 1);

-- Critical Chain Project Management → Project & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (11, 5);

-- Root Cause Analysis (DMAIC, Lean Six Sigma) → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (12, 2);

-- Value Stream Mapping → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (13, 2);

-- Influence & Persuasion Principles (Cialdini) → Leadership & Stakeholder Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (14, 6);

-- Monte Carlo Simulation → Risk Management & Decision Analysis
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (15, 7);

-- Dependency Mapping / PERT Analysis → Project & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (16, 5);

-- Galbraith Star Model → Organizational Design
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (17, 8);
```

```
-- Hoshin Kanri → Strategic Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (18, 1);

-- McKinsey 7S Framework → Organizational Design
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (19, 8);

-- RACI Matrix → Project & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (20, 5);

-- Portfolio Prioritization Matrix → Strategic Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (21, 1);

-- MoSCoW Prioritization → Project & Portfolio Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (22, 5);

-- Stage-Gate Process → Product & Portfolio Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (23, 9);

-- Scenario Analysis → Strategic & Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (24, 3);

-- Strategic Choice Cascade → Strategic Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (25, 1);

-- VRIO Framework → Strategic Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (26, 1);

-- PESTEL Analysis → Strategic Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (27, 1);

-- BCG Growth-Share Matrix → Portfolio Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (28, 9);

-- GE/McKinsey Matrix → Portfolio Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (29, 9);

-- Blue Ocean Strategy → Innovation & Market Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (30,
10);

-- Jobs to Be Done Framework → Product & Customer Insights
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (31,
11);
```

```
-- Value Proposition Canvas → Product & Customer Insights
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (32,
11);

-- Stakeholder Mapping → Leadership & Stakeholder Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (33, 6);

-- Customer Journey Mapping → Product & Marketing
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (34,
12);

-- AIDA / Pirate Metrics (AARRR) → Marketing & Growth
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (35,
13);

-- Conversion Funnel Analysis → Marketing & Growth
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (36,
13);

-- Net Promoter Score → Customer Experience
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (37,
14);

-- Customer Experience Maturity Model → Customer Experience
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (38,
14);

-- Brand Archetype Framework → Marketing & Branding
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (39,
13);

-- Marketing Mix / 4Ps → Marketing & Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (40,
13);

-- STP Model → Marketing & Product Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (41,
13);

-- Horizon 1-3 Innovation Model → Innovation & Portfolio
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (42,
10);
```

```
-- Innovation Ambition Matrix → Innovation & Portfolio
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (43, 10);

-- Disruptive Innovation Model → Innovation & Market
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (44, 10);

-- Product-Market Fit Pyramid → Product & Marketing
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (45, 11);

-- OKR Cascading Framework → Strategy & Performance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (46, 3);

-- Strategic Risk Assessment → Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (47, 7);
-- Batch 2: framework_knowledge_link (48–100)

-- Lean Six Sigma → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (48, 2);

-- DMAIC Cycle → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (49, 2);

-- Kaizen / Continuous Improvement → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (50, 2);

-- PDCA Cycle → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (51, 2);

-- Total Quality Management (TQM) → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (52, 2);

-- Just-in-Time (JIT) Production → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (53, 2);

-- Kanban → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (54, 2);

-- Scrum / Agile Framework → Project & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (55, 5);

-- Critical Path Method (CPM) → Project & Operations
```

```
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (56, 5);

-- PERT Analysis → Project & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (57, 5);

-- Bottleneck Analysis → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (58, 2);

-- SIPOC Mapping → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (59, 2);

-- Business Process Reengineering (BPR) → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (60, 2);

-- Process Capability Analysis (Cp, Cpk) → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (61, 2);

-- Failure Mode & Effects Analysis (FMEA) → Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (62, 7);

-- Service Blueprinting → Product & Customer Insights
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (63,
11);

-- Work Breakdown Structure (WBS) → Project & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (64, 5);

-- Resource Loading & Leveling → Project & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (65, 5);

-- Demand-Capacity Balancing → Project & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (66, 5);

-- Time-Driven Activity-Based Costing (TDABC) → Finance & Accounting
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (67,
15);

-- PMBOK Knowledge Areas → Project & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (68, 5);

-- PRINCE2 Methodology → Project & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (69, 5);

-- Stage-Gate Process → Product & Portfolio Management
```

```
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (70, 9);

-- RAID Log (Risks, Assumptions, Issues, Dependencies) → Project & Risk
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (71, 7);

-- RACI Matrix (Project Application) → Project & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (72, 5);

-- MoSCoW Prioritization → Project & Portfolio Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (73, 5);

-- Eisenhower Matrix → Productivity & Prioritization
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (74, 4);

-- Decision Matrix / Weighted Scoring → Project & Portfolio Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (75, 5);

-- Force Field Analysis → Change Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (76,
16);

-- Gap Analysis → Strategy & Performance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (77, 3);

-- Sensitivity Analysis → Risk Management & Decision Analysis
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (78, 7);

-- Pareto Analysis (80/20 Rule) → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (79, 2);

-- Ladder of Inference → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (80,
17);

-- Red Team / Blue Team Analysis → Risk Management & Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (81, 7);

-- Premortem Analysis → Risk Management & Project Planning
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (82, 7);

-- Influence Diagram → Decision Analysis
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (83,
17);
```

```
-- Causal Loop Diagram → Systems & Process Analysis
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (84, 2);

-- Systems Thinking Loops → Systems & Process Analysis
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (85, 2);

-- Heuristics & Bias Checklist → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (86,
17);

-- Decision Fatigue Awareness → Leadership & Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (87, 6);

-- Transactional Analysis → Leadership & Team Dynamics
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (88, 6);

-- Stress-Performance Curve (Yerkes-Dodson) → Leadership & Performance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (89, 6);

-- Flow State Model (Csikszentmihalyi) → Leadership & Performance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (90, 6);

-- Cognitive Bias Codex → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (91,
17);

-- Self-Determination Theory → Leadership & Motivation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (92, 6);

-- Emotional Intelligence (Goleman) → Leadership & Team Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (93, 6);

-- Situational Leadership (Hersey-Blanchard) → Leadership & Team Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (94, 6);

-- Servant Leadership Model → Leadership & Team Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (95, 6);

-- Kotter's 8-Step Change Model → Change Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (96,
16);

-- Lewin's Change Model → Change Management
```

```
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (97,
16);

-- ADKAR Model → Change Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (98,
16);

-- Tuckman Team Development Model → Leadership & Team Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (99, 6);

-- High-Performance Organization Model → Leadership & Organizational Performance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (100,
6);
-- Batch 3: framework_knowledge_link (101–150)

-- Talent Management Maturity Model → Leadership & HR
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (101,
6);

-- Leadership Pipeline → Leadership & Team Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (102,
6);

-- Cultural Web Model → Organizational Design & Culture
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (103,
6);

-- Decision Rights Matrix → Project & Organizational Governance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (104,
5);

-- Span of Control Analysis → Leadership & Organizational Design
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (105,
6);

-- Maslow's Hierarchy of Needs → Leadership & Motivation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (106,
6);

-- Herzberg Motivation-Hygiene → Leadership & Motivation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (107,
6);

-- McClelland's Theory of Needs → Leadership & HR
```

```
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (108, 6);

-- Self-Determination Theory → Leadership & Motivation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (109, 6);

-- Big Five Personality Traits (OCEAN) → Team & Individual Assessment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (110, 6);

-- DISC Assessment → Team & Individual Assessment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (111, 6);

-- Emotional Intelligence (Goleman) → Leadership & Team Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (112, 6);

-- Cognitive Load Theory → Project & Team Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (113, 5);

-- Behavioral Economics Principles → Decision-making & Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (114, 17);

-- Nudge Theory → Behavioral Interventions & Team
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (115, 17);

-- Loss Aversion / Prospect Theory → Risk Management & Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (116, 7);

-- Anchoring & Adjustment Bias → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (117, 17);

-- Availability Heuristic → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (118, 17);

-- Representativeness Heuristic → Decision-making
```

```
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (119, 17);

-- Sunk Cost Fallacy Awareness → Decision & Portfolio Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (120, 17);

-- Confirmation Bias Awareness → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (121, 17);

-- Framing Effect Awareness → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (122, 17);

-- Anchoring / Adjustment in Negotiation → Stakeholder & Influence
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (123, 6);

-- Social Proof & Bandwagon Effect → Team & Influence
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (124, 6);

-- Scarcity Principle → Resource & Stakeholder Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (125, 6);

-- Habit Loop (Cue-Routine-Reward) → Behavioral Interventions
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (126, 6);

-- Fogg Behavior Model → Behavioral Interventions
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (127, 6);

-- COM-B Behavior Model → Change & Behavioral Interventions
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (128, 16);

-- ADKAR Model → Change Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (129, 16);

-- Gantt Charts → Project & Operations
```

```
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (130, 5);

-- Monte Carlo Simulation → Risk & Investment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (131, 7);

-- Decision Tree Analysis → Risk & Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (132, 17);

-- Bayesian Updating → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (133, 17);

-- Risk Heat Map (ISO 31000) → Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (134, 7);

-- Enterprise Risk Management (ERM) → Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (135, 7);

-- Failure Mode & Effects Analysis (FMEA) → Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (136, 7);

-- Hazard & Operability Analysis (HAZOP) → Risk & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (137, 7);

-- Bowtie Risk Analysis → Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (138, 7);

-- COSO ERM Framework → Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (139, 7);

-- ISO 31000 Risk Management → Risk & Compliance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (140, 7);

-- Basel III / Banking Risk Framework → Financial Operations
```

```
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (141, 15);

-- CAPM (Capital Asset Pricing Model) → Finance & Investment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (142, 15);

-- Modern Portfolio Theory (MPT) → Finance & Investment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (143, 15);

-- Black-Litterman Model → Finance & Investment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (144, 15);

-- Risk-Adjusted Return on Capital (RAROC) → Finance & Risk
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (145, 15);

-- EVA / Economic Value Added → Finance & Performance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (146, 15);

-- Balanced Scorecard – Financial KPIs → Strategy & Finance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (147, 3);

-- Activity-Based Costing (ABC) → Finance & Accounting
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (148, 15);

-- Time-Driven ABC → Finance & Accounting
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (149, 15);

-- KPI Tree / Dashboard → Strategy & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (150, 3);
-- Batch 4: framework_knowledge_link (151–200)

-- Leading vs Lagging Indicators → Strategy & Performance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (151, 3);
```

```
-- Benchmarking (Competitive / Internal) → Strategy & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (152, 3);

-- Hoshin Kanri / Policy Deployment → Strategy & Planning
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (153, 3);

-- Organizational Network Analysis (ONA) → Team & Stakeholder Mapping
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (154, 6);

-- Sociomatrix / Social Graph Analysis → Team & Stakeholder Mapping
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (155, 6);

-- Complexity Theory Applications → Systems & Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (156, 3);

-- Systems Dynamics Modeling → Operations & Continuous Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (157, 2);

-- Feedback Loop Analysis → Operations & Systems Thinking
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (158, 2);

-- Control Charts → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (159, 2);

-- Statistical Process Control (SPC) → Operations & Quality
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (160, 2);

-- Six Sigma DMAIC → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (161, 2);

-- Six Sigma DMADV → Operations & Design
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (162, 2);
```

```
-- Design of Experiments (DOE) → Operations & Process Analysis
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (163, 2);

-- Taguchi Methods → Product Design & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (164, 2);

-- Root Cause Corrective Action (RCCA) → Post-event Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (165, 2);

-- Corrective & Preventive Actions (CAPA) → Post-event Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (166, 2);

-- Quality Function Deployment (QFD) → Product & Customer Design
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (167, 2);

-- Kano Model → Product & Customer Adoption
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (168, 2);

-- House of Quality → Product & Customer Design
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (169, 2);

-- Failure Tree Analysis (FTA) → Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (170, 7);

-- Event Tree Analysis (ETA) → Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (171, 7);

-- Probabilistic Risk Assessment (PRA) → Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (172, 7);

-- Decision Quality Framework → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (173, 17);
```

```
-- NPV / Net Present Value → Finance & Investment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (174, 15);

-- IRR / Internal Rate of Return → Finance & Investment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (175, 15);

-- Payback Period Analysis → Finance & Investment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (176, 15);

-- Risk-Adjusted ROI → Finance & Investment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (177, 15);

-- Real Options Analysis → Finance & Investment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (178, 15);

-- Monte Carlo Simulation for Finance → Risk & Investment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (179, 15);

-- TRIZ (Theory of Inventive Problem Solving) → Innovation & Problem-solving
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (180, 8);

-- Design Thinking → Product & Innovation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (181, 8);

-- Double Diamond Model → Product & Innovation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (182, 8);

-- Human-Centered Design (HCD) → Product & UX
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (183, 8);

-- Jobs-to-Be-Done (JTBD) → Product & Market
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (184, 8);
```

```
-- Lean Startup / Build-Measure-Learn → Product & MVP
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (185, 8);

-- Innovation Ambition Matrix → Innovation & Portfolio
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (186, 8);

-- Horizon 1-3 Innovation Model → Innovation & Portfolio
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (187, 8);

-- Blue Ocean Strategy → Innovation & Market
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (188, 8);

-- Disruptive Innovation Model → Innovation & Market
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (189, 8);

-- Technology Adoption Lifecycle → Product & Market
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (190, 8);

-- Rogers Diffusion of Innovation → Market & Adoption
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (191, 8);

-- Open Innovation Model → R&D & Collaboration
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (192, 8);

-- TRIZ 40 Principles → Innovation & Problem-solving
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (193, 8);

-- Morphological Analysis → Problem-solving & Scenario Planning
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (194, 8);

-- Brainstorming Techniques → Team & Innovation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (195, 8);
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-- SCAMPER Technique → Product Innovation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (196, 8);

-- Delphi Method → Strategic Forecasting
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (197, 8);

-- Nominal Group Technique → Team & Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (198, 8);

-- Futures Wheel → Scenario Planning
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (199, 8);

-- Backcasting → Strategic Planning
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (200, 8);
-- Batch 5: framework_knowledge_link (201–250)

-- Monte Carlo Simulation – Advanced → Risk & Investment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (201, 15);

-- Bayesian Decision Networks → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (202, 17);

-- Influence Diagram – Advanced → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (203, 17);

-- Game Theory / Nash Equilibrium → Competitive Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (204, 3);

-- Prisoner’s Dilemma Analysis → Competitive Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (205, 3);

-- Zero-Sum vs Non-Zero-Sum Analysis → Competitive Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (206, 3);

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-- Payoff Matrix → Competitive / Initiative Planning
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (207, 3);

-- Decision Quality Framework → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (208, 17);

-- Multi-Criteria Decision Analysis (MCDA) → Portfolio / Prioritization
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (209, 3);

-- Analytic Hierarchy Process (AHP) → Portfolio / Prioritization
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (210, 3);

-- Analytic Network Process (ANP) → Portfolio / Dependency
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (211, 3);

-- Weighted Scoring Model → Portfolio Prioritization
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (212, 3);

-- Cost-Benefit Analysis → Investment / Decision
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (213, 15);

-- Systems Thinking → Operations & Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (214, 2);

-- Feedback Loop Mapping → Operations / Continuous Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (215, 2);

-- Causal Loop Diagram → Operations / Continuous Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (216, 2);

-- Stock & Flow Modeling → Operations / System Dynamics
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (217, 2);
```

-- Organizational Network Analysis (ONA) – Advanced → Team & Stakeholder Mapping
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (218, 6);

-- Social Network Analysis → Team & Stakeholder Mapping
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (219, 6);

-- Bayesian Inference for Risk → Risk & Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (220, 7);

-- Predictive Analytics / Regression Modeling → Operations & KPI Analytics
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (221, 2);

-- Prescriptive Analytics → Operations & Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (222, 2);

-- Operations Research (OR) Models → Resource & Process Optimization
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (223, 2);

-- Linear Programming (LP) → Resource Allocation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (224, 2);

-- Integer Programming (IP) → Resource Allocation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (225, 2);

-- Queuing Theory → Resource & Operational Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (226, 2);

-- Simulation Modeling → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (227, 2);

-- Theory of Constraints (Goldratt) – Advanced → Operations & Dependency Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (228, 2);

-- Bottleneck Analysis – Advanced → Operations & Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (229, 2);

-- Constraint Management Framework → Operations & Process Optimization
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (230, 2);

-- Operational Maturity Models (OMM) → Organizational Assessment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (231, 3);

-- Capability Maturity Model Integration (CMMI) → Organizational Assessment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (232, 3);

-- ITIL Framework → IT Operations & Governance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (233, 9);

-- COBIT Framework → IT Governance & Risk
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (234, 9);

-- PMO Maturity Model → Project Office & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (235, 5);

-- EFQM Excellence Model → Organizational Performance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (236, 3);

-- Malcolm Baldrige Excellence Framework → Organizational Performance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (237, 3);

-- Benchmarking & Best Practices → Continuous Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (238, 3);

-- ISO Standards (9001, 14001, 45001) → Quality & Compliance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (239, 9);

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-- Cross-Industry Operational Frameworks → Strategy & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (240, 3);

-- Maslow's Hierarchy of Needs → Leadership & Motivation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (241, 6);

-- Herzberg Motivation-Hygiene → Leadership & Team
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (242, 6);

-- McClelland's Theory of Needs → Leadership & Talent Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (243, 6);

-- Self-Determination Theory → Leadership & Motivation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (244, 6);

-- Big Five Personality Traits (OCEAN) → Team & Individual Assessment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (245, 6);

-- DISC Assessment → Leadership & Team Behavior
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (246, 6);

-- Emotional Intelligence (Goleman) → Leadership & Team Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (247, 6);

-- Cognitive Load Theory → Decision & Project Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (248, 17);

-- Behavioral Economics Principles → Decision & Advisory
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (249, 17);

-- Nudge Theory → Decision & Behavior
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (250, 17);
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-- Batch 6: framework_knowledge_link (251–end)

-- Loss Aversion / Prospect Theory → Risk & Decision
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (251, 17);

-- Anchoring & Adjustment Bias → Decision-making / Heuristics
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (252, 17);

-- Availability Heuristic → Decision-making / Heuristics
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (253, 17);

-- Representativeness Heuristic → Decision-making / Heuristics
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (254, 17);

-- Sunk Cost Fallacy Awareness → Portfolio & Initiative Decisions
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (255, 3);

-- Confirmation Bias Awareness → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (256, 17);

-- Framing Effect Awareness → Decision-making / Behavioral Economics
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (257, 17);

-- Anchoring / Adjustment in Negotiation → Stakeholder & Negotiation Advisory
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (258, 6);

-- Social Proof & Bandwagon Effect → Team & Influence
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (259, 6);

-- Scarcity Principle → Resource Allocation / Advisory
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (260, 3);

-- Habit Loop (Cue-Routine-Reward) → Behavioral Interventions

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INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (261, 6);

-- Fogg Behavior Model → Team & Behavioral Interventions
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (262, 6);

-- COM-B Behavior Model → Change & Team Advisory
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (263, 6);

-- ADKAR Model → Change Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (264, 6);

-- Gantt Charts → Project Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (265, 5);

-- Monte Carlo Simulation → Risk & Initiative
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (266, 15);

-- Decision Tree Analysis → Initiative / Risk
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (267, 17);

-- Bayesian Updating → Decision-making
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (268, 17);

-- Risk Heat Map (ISO 31000) → Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (269, 7);

-- Enterprise Risk Management (ERM) → Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (270, 7);

-- Failure Mode & Effects Analysis (FMEA) → Risk & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (271, 7);

-- Hazard & Operability Analysis (HAZOP) → Operational Risk
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INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (272, 7);

-- Bowtie Risk Analysis → Risk Management
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (273, 7);

-- COSO ERM Framework → Risk & Compliance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (274, 7);

-- ISO 31000 Risk Management → Risk & Compliance
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (275, 7);

-- Basel III / Banking Risk Framework → Financial Risk
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (276, 15);

-- CAPM (Capital Asset Pricing Model) → Finance / Investments
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (277, 15);

-- Modern Portfolio Theory (MPT) → Finance / Portfolio
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (278, 15);

-- Black-Litterman Model → Portfolio Optimization
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (279, 15);

-- Risk-Adjusted Return on Capital (RAROC) → Finance / Risk
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (280, 15);

-- EVA / Economic Value Added → Finance / KPI
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (281, 15);

-- Balanced Scorecard – Financial KPIs → Finance / Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (282, 3);

-- Activity-Based Costing (ABC) → Finance / Costing
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INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (283, 15);

-- Time-Driven ABC → Finance / Costing
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (284, 15);

-- KPI Tree / Dashboard → Operations & Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (285, 2);

-- Leading vs Lagging Indicators → KPI & Reporting
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (286, 2);

-- Benchmarking (Competitive / Internal) → Strategy & Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (287, 3);

-- Hoshin Kanri / Policy Deployment → Strategic Planning
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (288, 3);

-- Organizational Network Analysis (ONA) – Advanced → Team & Stakeholder
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (289, 6);

-- Sociomatrix / Social Graph Analysis → Team & Stakeholder Mapping
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (290, 6);

-- Complexity Theory Applications → Strategy & Systems
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (291, 3);

-- Systems Dynamics Modeling → Operations / Continuous Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (292, 2);

-- Feedback Loop Analysis → Continuous Operations
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (293, 2);

-- Control Charts → Quality Management
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INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (294, 9);

-- Statistical Process Control (SPC) → Quality / Lean
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (295, 9);

-- Six Sigma DMAIC → Process Improvement
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (296, 9);

-- Six Sigma DMADV → Process / Product Design
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (297, 9);

-- Design of Experiments (DOE) → Product / Process Optimization
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (298, 9);

-- Taguchi Methods → Product / Design Robustness
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (299, 9);

-- Root Cause Corrective Action (RCCA) → Post-event / Process Fix
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (300, 2);

-- Corrective & Preventive Actions (CAPA) → Post-event / Process Fix
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (301, 2);

-- Quality Function Deployment (QFD) → Product / Customer Voice
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (302, 9);

-- Kano Model → Product / Customer
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (303, 9);

-- House of Quality → Product / Customer Requirements
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (304, 9);

-- Failure Tree Analysis (FTA) → Risk / Product
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INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (305, 7);

-- Event Tree Analysis (ETA) → Risk / Product
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (306, 7);

-- Probabilistic Risk Assessment (PRA) → Risk / Initiative
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (307, 7);

-- Decision Quality Framework → Decision & Advisory
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (308, 17);

-- NPV / Net Present Value → Investment Analysis
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (309, 15);

-- IRR / Internal Rate of Return → Investment Analysis
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (310, 15);

-- Payback Period Analysis → Investment Analysis
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (311, 15);

-- Risk-Adjusted ROI → Investment & Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (312, 15);

-- Real Options Analysis → Investment & Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (313, 15);

-- Monte Carlo Simulation for Finance → Investment & Risk
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (314, 15);

-- TRIZ (Theory of Inventive Problem Solving) → Innovation / Product
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (315, 3);

-- Design Thinking → Product / Innovation
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INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (316, 3);

-- Double Diamond Model → Product / Innovation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (317, 3);

-- Human-Centered Design (HCD) → Product / UX
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (318, 3);

-- Jobs-to-Be-Done (JTBD) → Product / Market
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (319, 3);

-- Lean Startup / Build-Measure-Learn → Product / MVP
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (320, 3);

-- Innovation Ambition Matrix → Innovation / Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (321, 3);

-- Horizon 1-3 Innovation Model → Innovation / Portfolio
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (322, 3);

-- Blue Ocean Strategy → Market Differentiation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (323, 3);

-- Disruptive Innovation Model → Market Strategy
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (324, 3);

-- Technology Adoption Lifecycle → Product / Market
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (325, 3);

-- Rogers Diffusion of Innovation → Market / Adoption
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (326, 3);

-- Open Innovation Model → R&D / Collaboration
```

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INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (327, 3);

-- TRIZ 40 Principles → Problem Solving / Innovation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (328, 3);

-- Morphological Analysis → Problem Solving / Scenarios
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (329, 3);

-- Brainstorming Techniques → Team / Innovation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (330, 6);

-- SCAMPER Technique → Product / Innovation
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (331, 3);

-- Delphi Method → Strategic Forecasting
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (332, 3);

-- Nominal Group Technique → Team Prioritization
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (333, 6);

-- Futures Wheel → Scenario Planning
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (334, 3);

-- Backcasting → Strategic Planning
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (335, 3);

-- Monte Carlo Simulation – Advanced → Risk / Investment
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (336, 15);

-- Bayesian Decision Networks → Decision-making / Probability
INSERT INTO framework_knowledge_link (framework_id, knowledge_base_id) VALUES (337, 17);
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